CA Risk Monitor

Sachu Mon P. Sajeev · TSI 2026

Dashboard Mode

- Real Data — VitalDB
- Synthetic — Deep Dive

VitalDB Patients

View Mode

- Single Patient
- Compare 2 Patients

Filter by

All Patients

Try fetch live waveform (VitalDB API)

Alert threshold (%)

55

Lookback window (min)

60

TAN · AUROC 0.9937 LightGBM · AUROC 0.9073 TSI University · 2026

INTRAOPERATIVE CARDIAC ARREST RISK MONITOR

REAL DATA — VitalDB

VitalDB · 6376 Patients · CA=66 · No-CA=6310 | 12 Apr 2026 · 14:13 | Compare 2 Patients

Sachu Mon P. Sajeev · MSc Data Science & AI · TSI University Riga · 2026

TAN · AUROC	LightGBM · AUROC	TAN · AUPRC
0.9937	0.9073	0.6027
95% CI [0.9909–0.9961]	60-min window	240-min window

Total: 6376 | CA: 66 | No CA: 6310 | 240min: 1731 | 120min: 4197 | 60min: 6047 | 30min: 6376

Patient A: #146 · CA · 240min

Patient B: #2 · NoCA · 240min

RISK COMPARISON SUMMARY

Case #146 · ⚠ CA EVENT · 240min

85.5%

CRITICAL

Case #2 · ✓ NO CA · 240min

9.2%

LOW

Case #146

⚠ CA EVENT · 240-min window · Feature-Reconstructed

✓ 30min · ✓ 60min · ✓ 120min · ✓ 240min

CA RISK · 240MIN WINDOW

85.3%

CRITICAL

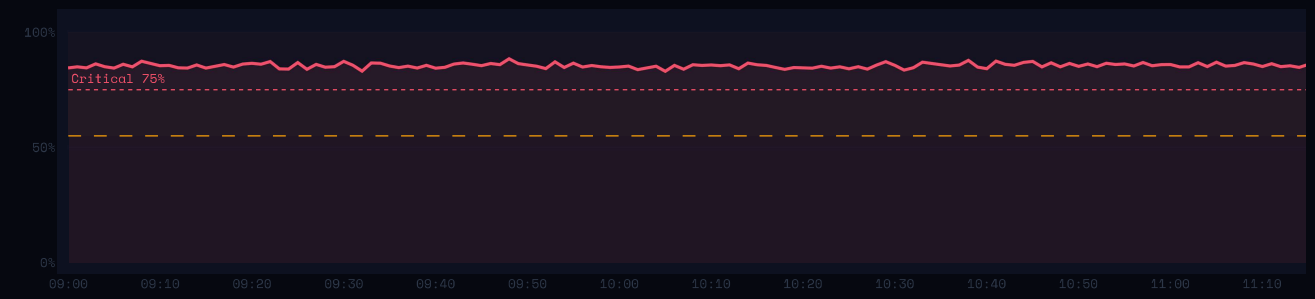
HEART RATE	SPO₂	ETCO₂	MEAN BP	SYSTOLIC BP	DIASTOLIC BP
88	100	35	115	128	106
bpm	%	mmHg	mmHg	mmHg	mmHg
→ OK	→ OK	↓ LOW	↑ HIGH	→ OK	↑ HIGH
→1.1/30m	→0.2/30m	→0.4/30m	↘16.8/30m	↘15.4/30m	↘17.8/30m
Range: 84–119	Range: 98–100	Range: 32–41	Range: -61–239	Range: -61–236	Range: -94–236
Normal: 50–100	Normal: 95–100	Normal: 35–45	Normal: 70–105	Normal: 90–140	Normal: 60–90

ALERT
Risk ≥ 55%
Notify physician

CLINICAL FEATURE STATISTICS (FROM CSV)

SIGNAL	MEAN	STD	MIN	MAX	SLOPE	NORMAL RANGE
Heart Rate	89.0	6.76	84.0	119.0	0.0012	50–100 bpm
SpO_2	99.8	0.41	98.0	100.0	-0.0001	95–100 %
$EtCO_2$	35.1	2.32	32.0	41.0	0.0004	35–45 mmHg
Mean BP	138.6	103.05	0.0	239.0	-0.0221	70–105 mmHg
Systolic BP	149.5	94.30	0.0	236.0	-0.0203	90–140 mmHg
Diastolic BP	131.0	108.88	0.0	236.0	-0.0235	60–90 mmHg

RISK SCORE OVER TIME



VITAL SIGNAL TRENDS

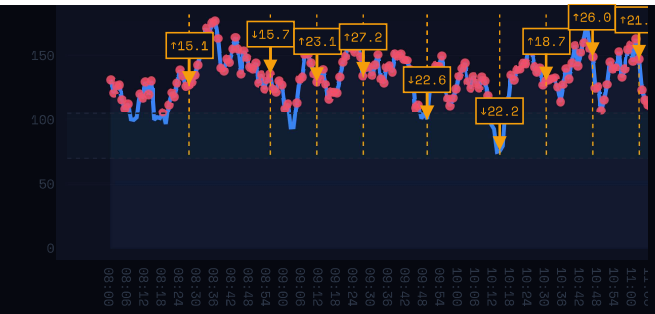
📊 Waveform reconstructed from CSV statistical features (mean, std, slope). VitalDB API fetch attempted automatically.

Heart Rate (bpm) · normal 50–100

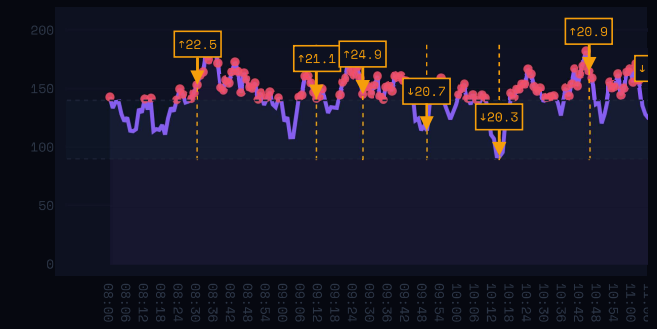
SpO_2 (%) · normal 95–100

$EtCO_2$ (mmHg) · normal 35–45

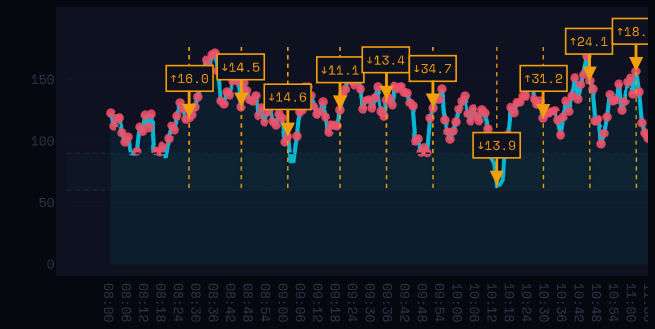
Mean BP (mmHg) · normal 70–105



Systolic BP (mmHg) · normal 90–140



Diastolic BP (mmHg) · normal 60–90

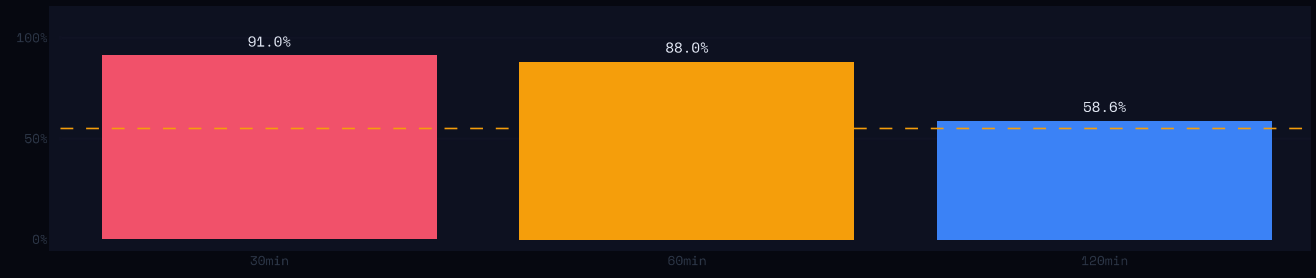


CHANGE POINT LOG

TIME	SIGNAL	VALUE	CHANGE	STATUS	NORMAL
08:27	Mean BP	126.9 mmHg	↑ 15.1	⚠ Outside range	70–105
08:27	Diastolic BP	118.6 mmHg	↑ 16.0	⚠ Outside range	60–90
08:30	Systolic BP	153.4 mmHg	↑ 22.5	⚠ Outside range	90–140
08:45	Diastolic BP	127.7 mmHg	↓ 14.5	⚠ Outside range	60–90
08:55	Mean BP	135.7 mmHg	↓ 15.7	⚠ Outside range	70–105
09:01	Diastolic BP	103.3 mmHg	↓ 14.6	⚠ Outside range	60–90
09:11	Mean BP	129.6 mmHg	↑ 23.1	⚠ Outside range	70–105
09:11	Systolic BP	141.3 mmHg	↑ 21.1	⚠ Outside range	90–140
09:19	Diastolic BP	125.3 mmHg	↓ 11.1	⚠ Outside range	60–90
09:27	Mean BP	133.9 mmHg	↑ 27.2	⚠ Outside range	70–105
09:27	Systolic BP	145.2 mmHg	↑ 24.9	⚠ Outside range	90–140
09:35	Diastolic BP	133.2 mmHg	↓ 13.4	⚠ Outside range	60–90
09:49	Mean BP	100.0 mmHg	↓ 22.6	Within range	70–105

TIME	SIGNAL	VALUE	CHANGE	STATUS	NORMAL
09:49	Systolic BP	114.2 mmHg	↓ 20.7	Within range	90–140
09:51	Diastolic BP	127.0 mmHg	↓ 34.7	⚠ Outside range	60–90
10:13	Diastolic BP	64.2 mmHg	↓ 13.9	Within range	60–90
10:14	Mean BP	76.1 mmHg	↓ 22.2	Within range	70–105
10:14	Systolic BP	92.3 mmHg	↓ 20.3	Within range	90–140
10:29	Diastolic BP	118.5 mmHg	↑ 31.2	⚠ Outside range	60–90
10:30	Mean BP	130.4 mmHg	↑ 18.7	⚠ Outside range	70–105
10:45	Systolic BP	164.9 mmHg	↑ 20.9	⚠ Outside range	90–140
10:45	Diastolic BP	148.7 mmHg	↑ 24.1	⚠ Outside range	60–90
10:46	Mean BP	149.1 mmHg	↑ 26.0	⚠ Outside range	70–105
11:01	Diastolic BP	156.6 mmHg	↑ 18.1	⚠ Outside range	60–90
11:02	Mean BP	147.2 mmHg	↑ 21.5	⚠ Outside range	70–105
11:09	Systolic BP	133.1 mmHg	↓ 26.2	Within range	90–140
11:17	Diastolic BP	138.3 mmHg	↑ 15.3	⚠ Outside range	60–90
11:18	Mean BP	147.9 mmHg	↑ 19.4	⚠ Outside range	70–105
11:33	Diastolic BP	136.1 mmHg	↑ 18.7	⚠ Outside range	60–90
11:43	Mean BP	126.5 mmHg	↓ 18.8	⚠ Outside range	70–105
11:51	Diastolic BP	124.4 mmHg	↑ 11.9	⚠ Outside range	60–90
11:59	Mean BP	114.8 mmHg	↓ 21.8	⚠ Outside range	70–105

RISK ACROSS AVAILABLE WINDOWS



Case #2 ✓ NO CA 240-min window 📊 Feature-Reconstructed

✓ 30min ✓ 60min ✓ 120min ✓ 240min

CA RISK · 240MIN WINDOW

9.0 %

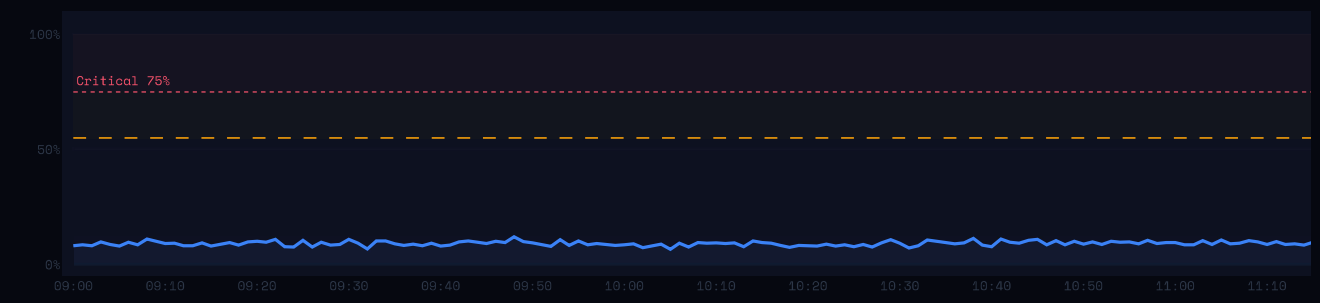
LOW

HEART RATE	SPO₂	ETCO₂	MEAN BP	SYSTOLIC BP	DIASTOLIC BP
74	100	35	87	114	75
bpm	%	mmHg	mmHg	mmHg	mmHg
→ OK	→ OK	→ OK	→ OK	→ OK	→ OK
→0.3/30m	→0.1/30m	→0.1/30m	→0.2/30m	→0.3/30m	→0.2/30m
Range: 69–81	Range: 100–100	Range: 32–38	Range: 0–0	Range: 0–0	Range: 0–0
Normal: 50–100	Normal: 95–100	Normal: 35–45	Normal: 70–105	Normal: 90–140	Normal: 60–90

CLINICAL FEATURE STATISTICS (FROM CSV)

SIGNAL	MEAN	STD	MIN	MAX	SLOPE	NORMAL RANGE
Heart Rate	75.0	5.15	69.0	81.0	0.0011	50–100 bpm
SpO_2	100.0	0.00	100.0	100.0	0.0000	95–100 %
$EtCO_2$	35.4	1.41	32.0	38.0	0.0000	35–45 mmHg
Mean BP	0.0	0.00	0.0	0.0	0.0000	70–105 mmHg
Systolic BP	0.0	0.00	0.0	0.0	0.0000	90–140 mmHg
Diastolic BP	0.0	0.00	0.0	0.0	0.0000	60–90 mmHg

RISK SCORE OVER TIME



VITAL SIGNAL TRENDS

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Heart Rate (bpm) · normal 50–100

SpO_2 (%) · normal 95–100

EtCO₂ (mmHg) · normal 35–45

Mean BP (mmHg) · normal 70–105

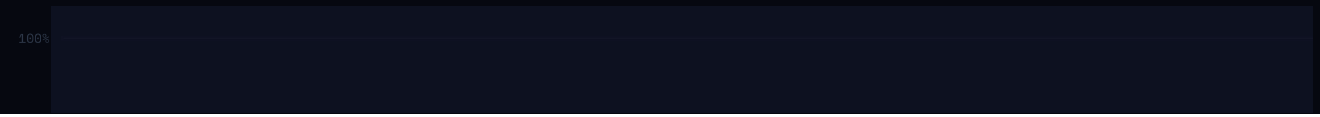
Systolic BP (mmHg) · normal 90–140

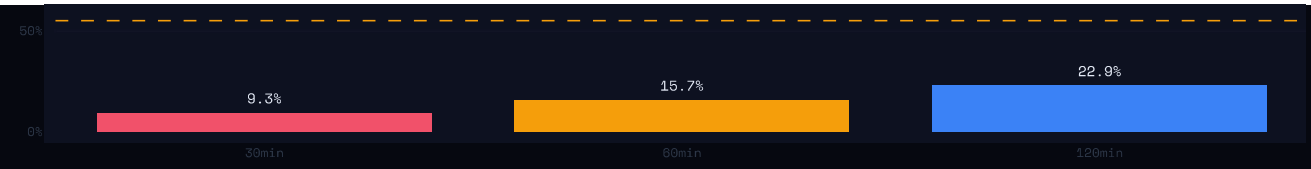
Diastolic BP (mmHg) · normal 60–90

CHANGE POINT LOG

No significant change points detected.

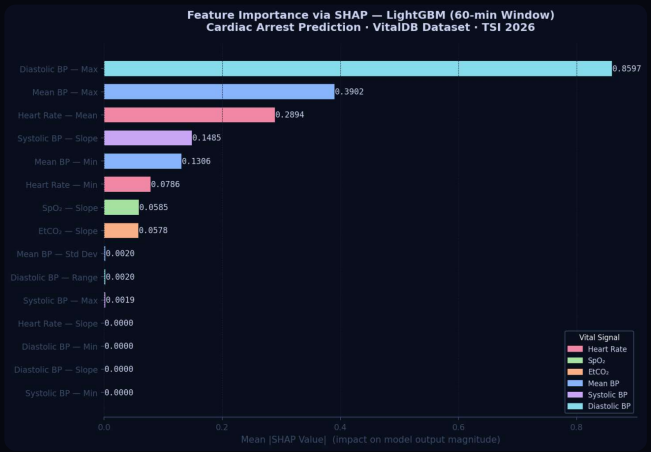
RISK ACROSS AVAILABLE WINDOWS



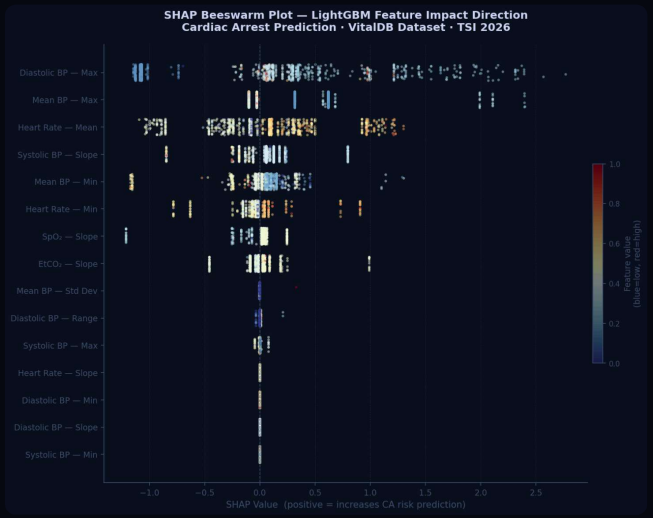


🔬 SHAP Feature Interpretability — LightGBM (60-min Window)

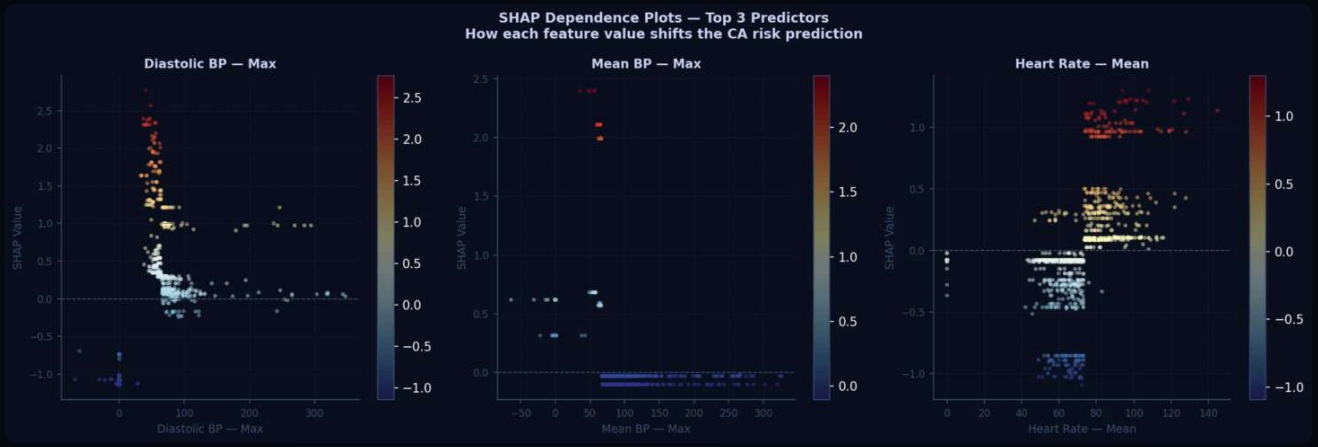
SHapley Additive exPlanations · TreeExplainer · AUROC 0.9073 · VitalDB Dataset · TSI 2026



Mean |SHAP| — Top 15 Features



SHAP Beeswarm — Feature Impact Direction



SHAP Dependence — Top 3 predictors

🔬 **How to read these charts:**

Bar chart — average magnitude of each feature's impact. Longer bar = stronger predictor.

Beeswarm — each dot is one patient. Red dots pushed right = high value increases CA risk. Blue pushed left = decreases risk.

Key finding — Diastolic BP Max, Mean BP Max, and Heart Rate Mean are the top 3 predictors. Run `python shap_analysis.py` to regenerate.

🟢 **240min** — Full 4-hour window | 🔵 **120min** — 2-hour window | 🟡 **60min** — 1-hour window | 🔴 **30min** — Minimal 30-min window

🔴 CRITICAL ≥75% — Alert physician immediately | 🟠 HIGH 50–75% — Increase monitoring | 🟡 MODERATE 25–50% — Watch closely | 🔵 LOW <25% — Routine monitoring

Clinical decision-support prototype · All treatment decisions remain with the attending physician · Sachu Mon P. Sajeev · MSc Data Science & AI · TSI University Riga · 2026